A four-field mimetic Arnold–Falk–Winther formulation with an explicit hydrostatic unknown

Abstract

We record the four-field form of the mimetic Arnold–Falk–Winther (AFW) discretisation of linear elasticity with weakly imposed symmetry, obtained by carrying the hydrostatic stress $p_s = \operatorname{tr}\boldsymbol{\sigma}/d$ as an independent cell unknown. The split is an exact change of variables, not an enrichment: the resulting system is congruent to the standard three-field one. Its purpose is structural. Coupled to Darcy flow, the Biot term degenerates from the discrete trace operator—which couples a cell pressure to every facet of that cell—to a diagonal cell–cell block. The deviatoric/hydrostatic separation also exposes the stress invariants I_1 and J_2 as primary unknowns.

1 Notation and the continuous problem

Let $\Omega \subset \mathbb{R}^d$, $d \in \{2, 3\}$, be a polytopal domain meshed by a complex whose d-cells are E and whose $(d-1)$-facets are e. Write $|E|$, $|e|$ for their measures, $\boldsymbol{n}_e$ for the facet normal in a fixed (canonical) orientation, and $s_{E,e} = \pm 1$ for the incidence sign of e in ∂E.

Elasticity in Hellinger–Reissner form with weakly imposed symmetry seeks a (not *a priori* symmetric) stress $\boldsymbol{\sigma}$, a displacement $\boldsymbol{u}$ and a rotation multiplier $\boldsymbol{s}$ with

$$\mathbb{C}^{-1}\boldsymbol{\sigma} = \nabla\boldsymbol{u} - \boldsymbol{s}, \qquad \operatorname{div}\boldsymbol{\sigma} = \boldsymbol{f}, \qquad \operatorname{skw}\boldsymbol{\sigma} = 0, \tag{1}$$

the last imposed weakly through $\boldsymbol{s}$. For isotropic material,

$$\mathbb{C}^{-1}\boldsymbol{\sigma} = \frac{1}{2\mu}\Big(\boldsymbol{\sigma} - a\ \operatorname{tr}(\boldsymbol{\sigma})\,\boldsymbol{I}\Big), \qquad a = \frac{\nu}{1 - 2\nu + d\nu}, \tag{2}$$

so that $a \to 1/d$ as $\nu \to \frac{1}{2}$. Note a is bounded on the whole range of ν; the incompressible limit degenerates only through $1 - ad$.

2 The deviatoric/hydrostatic split

Set

$$p_s := \tfrac{1}{d}\operatorname{tr}\boldsymbol{\sigma}, \qquad \boldsymbol{\sigma}_{\mathrm{dev}} := \boldsymbol{\sigma} - p_s\boldsymbol{I}, \qquad \operatorname{tr}\boldsymbol{\sigma}_{\mathrm{dev}} = 0. \tag{3}$$

The complementary energy separates exactly:

$$\boldsymbol{\sigma} : \mathbb{C}^{-1}\boldsymbol{\sigma} = \underbrace{\frac{1}{2\mu}\,\boldsymbol{\sigma}_{\mathrm{dev}} : \boldsymbol{\sigma}_{\mathrm{dev}}}_{\text{deviatoric}} + \underbrace{\frac{d\,(1 - ad)}{2\mu}\,p_s^2}_{\text{hydrostatic}}. \tag{4}$$

The two terms involve disjoint variables. This is the algebraic content of the split: the compliance is block diagonal in $(\boldsymbol{\sigma}_{\mathrm{dev}}, p_s)$.

Remark 1. The hydrostatic coefficient $d(1 - ad)/2\mu$ vanishes as $\nu \to \frac{1}{2}$, which is the incompressibility constraint appearing as a vanishing diagonal rather than as an unbounded one. This is the reason to carry p_s explicitly.

3 Discrete spaces

Facet stress degrees of freedom are traction moments; write n_{df} for their number per facet. Two choices are used here:

space	n_{df}	consistency modes	exactness
AFW	d^2	$d^2(d+1)$	constants and linears, any mesh
lumped	d	d^2	constants, orthogonal cells only

The hydrostatic unknown p_s is one scalar per cell, the displacement $\boldsymbol{u}$ is d per cell and the rotation $\boldsymbol{s}$ is $d(d-1)/2$ per cell.

4 Operators

Let D be the discrete divergence and A the discrete asymmetry, both acting on facet stress degrees of freedom, and let M_{dev} be the inner product induced by the first term of (**??**). Define the *hydrostatic embedding* E, mapping a cell scalar to the facet tractions of $p_s\boldsymbol{I}$,

$$(E\,p_s)_e \;=\; s_{E,e}\,|e|\,p_s\,\boldsymbol{n}_e, \tag{5}$$

and the diagonal hydrostatic inner product

$$(M_{\text{vol}})_{EE} = |E|\,\frac{d\,(1-ad)}{2\mu}. \tag{6}$$

Proposition 1 (Structure of the split operators)**.** *With E as in* (**??**),

$$D_{\text{vol}} = D\,E, \tag{7}$$
$$A\,E = 0. \tag{8}$$

Proof. (**??**): the divergence acts on the traction, and E is by definition the traction of $p_s\boldsymbol{I}$; there is no second divergence operator, only the embedding. (**??**): $\text{skw}(p_s\boldsymbol{I}) = 0$ because $\boldsymbol{I}$ is symmetric, so the hydrostatic part carries no asymmetry. □

Identity (**??**) is what keeps D purely topological: it remains the signed incidence matrix, composed with the diagonal metric factor $|e|\boldsymbol{n}_e$ carried by E. Identity (**??**) has a consequence developed in Section **??**.

5 The four-field system

Elasticity alone reads

$$\begin{bmatrix} M_{\text{dev}} & 0 & D^\top & A^\top \\ 0 & M_{\text{vol}} & D_{\text{vol}}^\top & 0 \\ D & D_{\text{vol}} & 0 & 0 \\ A & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} \boldsymbol{\sigma}_{\text{dev}} \\ p_s \\ \boldsymbol{u} \\ \boldsymbol{s} \end{bmatrix} = \begin{bmatrix} \boldsymbol{g} \\ 0 \\ \boldsymbol{f} \\ 0 \end{bmatrix}. \tag{9}$$

5.1 Equivalence with the three-field system

Let T be the change of variables $\boldsymbol{\sigma} = \boldsymbol{\sigma}_{\text{dev}} + E\,p_s$, i.e.

$$T = \begin{bmatrix} I & -E & 0 & 0 \\ 0 & I & 0 & 0 \\ 0 & 0 & I & 0 \\ 0 & 0 & 0 & I \end{bmatrix}. \tag{10}$$

Applying the congruence $T^\top K\,T$ to (??) and using (??) and (??)—which annihilate the $(3,2)$ and $(4,2)$ blocks exactly—gives

$$\begin{bmatrix} M_{\text{dev}} & -M_{\text{dev}}E & D^\top & A^\top \\ -E^\top M_{\text{dev}} & E^\top M_{\text{dev}}E + M_{\text{vol}} & 0 & 0 \\ D & 0 & 0 & 0 \\ A & 0 & 0 & 0 \end{bmatrix}, \tag{11}$$

whose $\boldsymbol{u}$ and $\boldsymbol{s}$ rows are $D\boldsymbol{\sigma} = \boldsymbol{f}$ and $A\boldsymbol{\sigma} = 0$. Eliminating p_s recovers the three-field system with

$$M_{\text{eff}} = M_{\text{dev}} - M_{\text{dev}}E\big(E^\top M_{\text{dev}}E + M_{\text{vol}}\big)^{-1}E^\top M_{\text{dev}}, \tag{12}$$

which is the full compliance in Woodbury form: M_{dev} plus a correction of rank one *per cell.* Eliminating p_s *before* the change of variables instead leaves M_{dev} untouched and produces a grad–div term $G = D_{\text{vol}}M_{\text{vol}}^{-1}D_{\text{vol}}^\top$ in the $(\boldsymbol{u},\boldsymbol{u})$ block. The three forms are related by congruence and share a solution; they differ only in where the hydrostatic energy is stored.

Remark 2. M_{eff} should be applied, not assembled: forming (??) explicitly fills in globally, whereas the Woodbury application costs a diagonal solve plus an auxiliary system of size n_{cells}.

6 Coupling to Darcy flow

With Darcy flux $\boldsymbol{q}$ and fluid pressure p_f, backward Euler over Δt, and Biot coefficient α, the six-field system is

$$\begin{bmatrix} M_{\text{dev}} & 0 & D^\top & A^\top & 0 & 0 \\ 0 & M_{\text{vol}} & D_{\text{vol}}^\top & 0 & 0 & \alpha I \\ D & D_{\text{vol}} & 0 & 0 & 0 & 0 \\ A & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\Delta t\, M_q & \Delta t\, B^\top \\ 0 & \alpha I & 0 & 0 & \Delta t\, B & S|E| \end{bmatrix} \begin{bmatrix} \boldsymbol{\sigma}_{\text{dev}} \\ p_s \\ \boldsymbol{u} \\ \boldsymbol{s} \\ \boldsymbol{q} \\ p_f \end{bmatrix} = \begin{bmatrix} \boldsymbol{g} \\ 0 \\ \boldsymbol{f} \\ 0 \\ -\Delta t\, \boldsymbol{g}_p \\ r \end{bmatrix}. \tag{13}$$

This is the motivation for the split. In the three-field form the Biot term enters as $\big(\frac{\alpha}{dK}T\big)^\top p_f$, where the discrete trace T couples each cell pressure to *every facet degree of freedom of that cell.* With p_s explicit the same coupling is the diagonal block αI and T disappears from the system. Consequences:

- the (p_s, p_f) sub-block is 2×2 per cell, with M_{vol} and the storage $S|E|$ both diagonal;
- a pressure Schur complement can be formed on a variable that has a nonzero diagonal, which is what a CPR-type preconditioner requires;
- the stress invariants are primary unknowns, $I_1 = d\,p_s$ and $J_2 = \frac{1}{2}\boldsymbol{\sigma}_{\text{dev}} : \boldsymbol{\sigma}_{\text{dev}}$, so a yield function $F(I_1, J_2)$ and its Jacobian need no reconstruction through T.

For interface contact the alignment is only partial. Writing the facet traction $\boldsymbol{t} = \boldsymbol{\sigma}\boldsymbol{n}$,

$$t_N = \boldsymbol{n}\cdot\boldsymbol{\sigma}_{\text{dev}}\boldsymbol{n} + p_s, \qquad \boldsymbol{t}_T = \boldsymbol{\sigma}_{\text{dev}}\boldsymbol{n} - (\boldsymbol{n}\cdot\boldsymbol{\sigma}_{\text{dev}}\boldsymbol{n})\,\boldsymbol{n}, \tag{14}$$

since $p_s\boldsymbol{I}\boldsymbol{n} = p_s\boldsymbol{n}$ is purely normal. The tangential traction is therefore independent of p_s, while the normal traction mixes; the Coulomb condition $|\boldsymbol{t}_T| \le \mu|t_N|$ still couples the two.

7 Degree-of-freedom counts

Per cell, asymptotically, with ρ the facet-to-cell ratio (3/2 for triangles, 2 for tetrahedra and quadrilaterals, 3 for hexahedra):

family	$\boldsymbol{\sigma}_{\text{dev}}$	p_s	$\boldsymbol{u}$	$\boldsymbol{s}$	$\boldsymbol{q}$	p_f	total	vs. unsplit
triangles	6.0	1	2	1	1.5	1	12.5	+8.7%
tetrahedra	18.0	1	3	3	2.0	1	28.0	+3.7%
quadrilaterals	8.0	1	2	1	2.0	1	15.0	+7.1%
hexahedra	27.0	1	3	3	3.0	1	38.0	+2.7%

The split costs one unknown per cell. The stress block dominates: 48% of the unknowns in 2D and 64% in 3D.

8 Sparsity

M_{vol} is diagonal by construction for any space. M_{dev} is diagonal only in the lumped space and only on orthogonal cells; for AFW it is dense within each cell, because the density arises from consistency on the $d^2(d+1)$ modes rather than from the hydrostatic coupling. Removing the hydrostatic direction is a rank-one change and cannot alter that.

9 On stability of the lumped variant

By (**??**) the rotation constraint does not see p_s. Hence the inf–sup pairing between the stress space and $\boldsymbol{s}$ is unchanged by the split, and no rearrangement of the hydrostatic unknown can affect it. This is a statement about the *space*, not about the choice of inner product.

Numerically, on a sequence of uniform quadrilateral meshes, the smallest eigenvalue of the Schur complement in the multiplier mass norm behaves as

h	1/2	1/4	1/8	1/16
AFW	2.97	1.95	1.90	1.87
lumped, d per facet	1.38	0.320	0.0814	0.0205

The AFW constant is bounded below; the lumped one decays as $O(h^2)$ and is recovered only by adding a rotation-jump term in the $(\boldsymbol{s}, \boldsymbol{s})$ block, at the cost of the zero block. The four-field split does not change either column.